

## 21.8.16 DFSDM register map

The following table summarizes the DFSDM registers.

Table 108. DFSDM register map and reset values

| Offset      | Register name   | 31           | 30       | 29   | 28   | 27   | 26   | 25   | 24   | 23            | 22   | 21   | 20          | 19   | 18   | 17   | 16           | 15           | 14          | 13          | 12   | 11   | 10   | 9          | 8       | 7    | 6      | 5     | 4    | 3              | 2    | 1         | 0 |
|-------------|-----------------|--------------|----------|------|------|------|------|------|------|---------------|------|------|-------------|------|------|------|--------------|--------------|-------------|-------------|------|------|------|------------|---------|------|--------|-------|------|----------------|------|-----------|---|
| 0x00        | DFSDM_CH0CFGR1  | DFSDMEN      | CKOUTSRC | Res. | Res. | Res. | Res. | Res. | Res. | CKOUTDIV[7:0] |      |      |             |      |      |      |              | DATPACK[1:0] |             | DATMPX[1:0] |      | Res. | Res. | Res.       | CHINSEL | CHEN | CKABEN | SCDEN | Res. | SPICKSEL [1:0] |      | SITP[1:0] |   |
|             | reset value     | 0            | 0        |      |      |      |      |      |      | 0             | 0    | 0    | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           | 0    | 0    |      |            |         | 0    | 0      | 0     | 0    |                | 0    | 0         | 0 |
| 0x04        | DFSDM_CH0CFGR2  | OFFSET[23:0] |          |      |      |      |      |      |      |               |      |      |             |      |      |      |              |              |             |             |      |      |      | DTRBS[4:0] |         |      |        |       | Res. | Res.           | Res. |           |   |
|             | reset value     | 0            |          |      |      |      |      |      |      |               |      |      |             |      |      |      |              |              |             |             |      |      |      | 0          |         |      |        |       |      |                |      |           |   |
| 0x08        | DFSDM_CH0AWSCDR | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | AWFORD [1:0]  |      | Res. | AWFOSR[4:0] |      |      |      | BKSCD[3:0]   |              |             | Res.        | Res. | Res. | Res. | SCDT[7:0]  |         |      |        |       |      |                |      |           |   |
|             | reset value     |              |          |      |      |      |      |      |      | 0             | 0    |      | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           |      |      |      |            | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x0C        | DFSDM_CH0WDATR  | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res.        | Res. | Res. | Res. | WDATA[15:0]  |              |             |             |      |      |      |            |         |      |        |       |      |                |      |           |   |
|             | reset value     |              |          |      |      |      |      |      |      |               |      |      |             |      |      |      | 0            | 0            | 0           | 0           | 0    | 0    | 0    | 0          | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x10        | DFSDM_CH0DATINR | INDAT1[15:0] |          |      |      |      |      |      |      |               |      |      |             |      |      |      | INDAT0[15:0] |              |             |             |      |      |      |            |         |      |        |       |      |                |      |           |   |
|             | reset value     | 0            | 0        | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           | 0    | 0    | 0    | 0          | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x14 - 0x1C | Reserved        | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res.        | Res. | Res. | Res. | Res.         | Res.         | Res.        | Res.        | Res. | Res. | Res. | Res.       | Res.    | Res. | Res.   | Res.  | Res. | Res.           | Res. | Res.      |   |
| 0x20        | DFSDM_CH1CFGR1  | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res.        | Res. | Res. | Res. | DATPACK[1:0] |              | DATMPX[1:0] |             | Res. | Res. | Res. | Res.       | CHINSEL | CHEN | CKABEN | SCDEN | Res. | SPICKSEL [1:0] |      | SITP[1:0] |   |
|             | reset value     |              |          |      |      |      |      |      |      |               |      |      |             |      |      |      | 0            | 0            | 0           | 0           |      |      |      | 0          | 0       | 0    | 0      |       | 0    | 0              | 0    | 0         |   |
| 0x24        | DFSDM_CH1CFGR2  | OFFSET[23:0] |          |      |      |      |      |      |      |               |      |      |             |      |      |      |              |              |             |             |      |      |      | DTRBS[4:0] |         |      |        |       | Res. | Res.           | Res. |           |   |
|             | reset value     | 0            | 0        | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           | 0    | 0    | 0    | 0          | 0       | 0    | 0      | 0     |      |                |      |           |   |
| 0x28        | DFSDM_CH1AWSCDR | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | AWFORD[1:0]   |      | Res. | AWFOSR[4:0] |      |      |      | BKSCD[3:0]   |              |             | Res.        | Res. | Res. | Res. | SCDT[7:0]  |         |      |        |       |      |                |      |           |   |
|             | reset value     |              |          |      |      |      |      |      |      | 0             | 0    |      | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           |      |      |      |            | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x2C        | DFSDM_CH1WDATR  | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res.        | Res. | Res. | Res. | WDATA[15:0]  |              |             |             |      |      |      |            |         |      |        |       |      |                |      |           |   |
|             | reset value     |              |          |      |      |      |      |      |      |               |      |      |             |      |      |      | 0            | 0            | 0           | 0           | 0    | 0    | 0    | 0          | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x30        | DFSDM_CH1DATINR | INDAT1[15:0] |          |      |      |      |      |      |      |               |      |      |             |      |      |      | INDAT0[15:0] |              |             |             |      |      |      |            |         |      |        |       |      |                |      |           |   |
|             | reset value     | 0            | 0        | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0           | 0    | 0    | 0    | 0            | 0            | 0           | 0           | 0    | 0    | 0    | 0          | 0       | 0    | 0      | 0     | 0    | 0              | 0    | 0         |   |
| 0x34 - 0x3C | Reserved        | Res.         | Res.     | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res.        | Res. | Res. | Res. | Res.         | Res.         | Res.        | Res.        | Res. | Res. | Res. | Res.       | Res.    | Res. | Res.   | Res.  | Res. | Res.           | Res. | Res.      |   |

Table 108. DFSDM register map and reset values (continued)

| Offset      | Register name   | 31           | 30     | 29   | 28  | 27  | 26       | 25  | 24  | 23          | 22  | 21     | 20          | 19  | 18    | 17    | 16           | 15           | 14  | 13          | 12  | 11  | 10  | 9            | 8   | 7          | 6         | 5      | 4     | 3   | 2             | 1   | 0         |   |
|-------------|-----------------|--------------|--------|------|-----|-----|----------|-----|-----|-------------|-----|--------|-------------|-----|-------|-------|--------------|--------------|-----|-------------|-----|-----|-----|--------------|-----|------------|-----------|--------|-------|-----|---------------|-----|-----------|---|
| 0x40        | DFSDM_CH2CFGR1  | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | DATPACK[1:0] |     | DATMPX[1:0] |     | Res | Res | Res          | Res | CHINSEL    | CHEN      | CKABEN | SCDEN | Res | SPICKSEL[1:0] |     | SITP[1:0] |   |
|             | reset value     |              |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              | 0            | 0   | 0           | 0   |     |     |              |     | 0          | 0         | 0      | 0     |     | 0             | 0   | 0         | 0 |
| 0x44        | DFSDM_CH2CFGR2  | OFFSET[23:0] |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              |              |     |             |     |     |     |              |     | DTRBS[4:0] |           |        |       | Res |               | Res | Res       |   |
|             | reset value     | 0            | 0      | 0    | 0   | 0   | 0        | 0   | 0   | 0           | 0   | 0      | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     |     |               |     |           |   |
| 0x48        | DFSDM_CH2AWSCDR | Res          | Res    | Res  | Res | Res | Res      | Res | Res | AWFORD[1:0] |     | Res    | AWFOSR[4:0] |     |       |       | BKSCD[3:0]   |              |     | Res         | Res | Res | Res | Res          | Res | SCDT[7:0]  |           |        |       |     |               |     |           |   |
|             | reset value     |              |        |      |     |     |          |     |     | 0           | 0   |        | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x4C        | DFSDM_CH2WDATR  | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | WDATA[15:0]  |     |             |     |     |     |              |     |            |           |        |       |     |               |     |           |   |
|             | reset value     |              |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x50        | DFSDM_CH2DATINR | INDAT1[15:0] |        |      |     |     |          |     |     |             |     |        |             |     |       |       | INDAT0[15:0] |              |     |             |     |     |     |              |     |            |           |        |       |     |               |     |           |   |
|             | reset value     | 0            | 0      | 0    | 0   | 0   | 0        | 0   | 0   | 0           | 0   | 0      | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x54 - 0x5C | Reserved        | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | Res          | Res | Res         | Res | Res | Res | Res          | Res | Res        | Res       | Res    | Res   | Res | Res           | Res | Res       |   |
| 0x60        | DFSDM_CH3CFGR1  | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | DATPACK[1:0] |     | DATMPX[1:0] |     | Res | Res | Res          | Res | CHINSEL    | CHEN      | CKABEN | SCDEN | Res | SPICKSEL[1:0] |     | SITP[1:0] |   |
|             | reset value     |              |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              | 0            | 0   | 0           | 0   |     |     |              | 0   | 0          | 0         | 0      |       | 0   | 0             | 0   | 0         |   |
| 0x64        | DFSDM_CH3CFGR2  | OFFSET[23:0] |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              |              |     |             |     |     |     |              |     | DTRBS[4:0] |           |        |       | Res |               | Res | Res       |   |
|             | reset value     | 0            | 0      | 0    | 0   | 0   | 0        | 0   | 0   | 0           | 0   | 0      | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     |     |               |     |           |   |
| 0x68        | DFSDM_CH3AWSCDR | Res          | Res    | Res  | Res | Res | Res      | Res | Res | AWFORD[1:0] |     | Res    | AWFOSR[4:0] |     |       |       | BKSCD[3:0]   |              |     | Res         | Res | Res | Res | Res          | Res | Res        | SCDT[7:0] |        |       |     |               |     |           |   |
|             | reset value     |              |        |      |     |     |          |     |     | 0           | 0   |        | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x6C        | DFSDM_CH3WDATR  | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | WDATA[15:0]  |     |             |     |     |     |              |     |            |           |        |       |     |               |     |           |   |
|             | reset value     |              |        |      |     |     |          |     |     |             |     |        |             |     |       |       |              | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x70        | DFSDM_CH3DATINR | INDAT1[15:0] |        |      |     |     |          |     |     |             |     |        |             |     |       |       | INDAT0[15:0] |              |     |             |     |     |     |              |     |            |           |        |       |     |               |     |           |   |
|             | reset value     | 0            | 0      | 0    | 0   | 0   | 0        | 0   | 0   | 0           | 0   | 0      | 0           | 0   | 0     | 0     | 0            | 0            | 0   | 0           | 0   | 0   | 0   | 0            | 0   | 0          | 0         | 0      | 0     | 0   | 0             | 0   | 0         |   |
| 0x74 - 0xFC | Reserved        | Res          | Res    | Res  | Res | Res | Res      | Res | Res | Res         | Res | Res    | Res         | Res | Res   | Res   | Res          | Res          | Res | Res         | Res | Res | Res | Res          | Res | Res        | Res       | Res    | Res   | Res | Res           | Res | Res       |   |
| 0x100       | DFSDM_FLT0CR1   | Res          | AWFSEL | FAST | Res | Res | RCH[1:0] |     | Res | Res         | Res | RDMAEN |             | Res | RSYNC | RCONT | RSW START    | Res          | Res | JEXTEN[1:0] |     | Res | Res | JEXTSEL[2:0] |     | Res        | Res       | Res    | Res   | Res | Res           | Res | Res       |   |
|             | reset value     |              | 0      | 0    |     |     |          | 0   | 0   |             |     | 0      |             | 0   | 0     | 0     | 0            |              |     | 0           | 0   |     | 0   | 0            | 0   |            |           | 0      | 0     | 0   | 0             | 0   | 0         |   |

Table 108. DFSDM register map and reset values (continued)

| Offset | Register name    | 31          | 30   | 29   | 28   | 27           | 26   | 25        | 24   | 23   | 22   | 21   | 20   | 19            | 18   | 17   | 16   | 15   | 14   | 13   | 12   | 11            | 10   | 9    | 8    | 7    | 6         | 5          | 4            | 3             | 2          | 1      | 0      |       |
|--------|------------------|-------------|------|------|------|--------------|------|-----------|------|------|------|------|------|---------------|------|------|------|------|------|------|------|---------------|------|------|------|------|-----------|------------|--------------|---------------|------------|--------|--------|-------|
| 0x104  | DFSDM_FLT0CR2    | Res.        | Res. | Res. | Res. | Res.         | Res. | Res.      | Res. | Res. | Res. | Res. | Res. | AWDCH[3:0]    |      |      |      | Res. | Res. | Res. | Res. | EXCH[3:0]     |      |      |      | Res. | CKABIE    | SCDIE      | AWDIE        | ROVRIE        | JOVRIE     | REOCIE | JEOCIE |       |
|        | reset value      |             |      |      |      |              |      |           |      |      |      |      |      | 0             | 0    | 0    | 0    |      |      |      |      | 0             | 0    | 0    | 0    |      | 0         | 0          | 0            | 0             | 0          | 0      | 0      |       |
| 0x108  | DFSDM_FLT0ISR    | Res.        | Res. | Res. | Res. | SCDF[3:0]    |      |           |      | Res. | Res. | Res. | Res. | CKABF[3:0]    |      |      |      | Res. | RCIP | JCIP | Res. | Res.          | Res. | Res. | Res. | Res. | Res.      | Res.       | Res.         | AWDF          | ROVRF      | JOVRF  | REOCF  | JEOCF |
|        | reset value      |             |      |      |      | 0            | 0    | 0         | 0    |      |      |      |      | 1             | 1    | 1    | 1    |      | 0    | 0    |      |               |      |      |      |      |           |            | 0            | 0             | 0          | 0      | 0      | 0     |
| 0x10C  | DFSDM_FLT0ICR    | Res.        | Res. | Res. | Res. | CLRSCDF[3:0] |      |           |      | Res. | Res. | Res. | Res. | CLRCKABF[3:0] |      |      |      | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | Res.      | Res.       | Res.         | CLR ROVRF     | CLR JOVRF  | Res.   | Res.   |       |
|        | reset value      |             |      |      |      | 0            | 0    | 0         | 0    |      |      |      |      | 0             | 0    | 0    | 0    |      |      |      |      |               |      |      |      |      |           |            |              | 0             | 0          |        |        |       |
| 0x110  | DFSDM_FLT0JCHGR  | Res.        | Res. | Res. | Res. | Res.         | Res. | Res.      | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | Res.      | Res.       | Res.         | JCHG[3:0]     |            |        |        |       |
|        | reset value      |             |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      |      |      |      |           |            | 0            | 0             | 0          | 1      |        |       |
| 0x114  | DFSDM_FLT0FCR    | FORD[2:0]   |      |      | Res. | Res.         | Res. | FOSR[9:0] |      |      |      |      |      |               |      | Res. | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | IOSR[7:0] |            |              |               |            |        |        |       |
|        | reset value      | 0           | 0    | 0    |      |              |      | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    |      |      |      |               |      |      |      |      | 0         | 0          | 0            | 0             | 0          | 0      | 0      | 0     |
| 0x118  | DFSDM_FLT0JDATAR | JDATA[23:0] |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      |      | Res. | Res. | Res.      | Res.       | Res.         | Res.          | JDATA[1:0] |        |        |       |
|        | reset value      | 0           | 0    | 0    | 0    | 0            | 0    | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    |      |           |            |              |               |            | 0      | 0      |       |
| 0x11C  | DFSDM_FLT0RDATAR | RDATA[23:0] |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      | Res. | Res. | Res. | Res.      | RPEND      | Res.         | Res.          | RDATA[1:0] |        |        |       |
|        | reset value      | 0           | 0    | 0    | 0    | 0            | 0    | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    |      |           |            | 0            |               |            | 0      | 0      |       |
| 0x120  | DFSDM_FLT0AWHTR  | AWHT[23:0]  |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      | Res. | Res. | Res. | Res.      | BKAWH[3:0] |              |               |            |        |        |       |
|        | reset value      | 0           | 0    | 0    | 0    | 0            | 0    | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    |      |      |           | 0          | 0            | 0             | 0          | 0      |        |       |
| 0x124  | DFSDM_FLT0AWLTR  | AWLT[23:0]  |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      | Res. | Res. | Res. | Res.      | BKAWL[3:0] |              |               |            |        |        |       |
|        | reset value      | 0           | 0    | 0    | 0    | 0            | 0    | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    |      |      |           | 0          | 0            | 0             | 0          | 0      |        |       |
| 0x128  | DFSDM_FLT0AWSR   | Res.        | Res. | Res. | Res. | Res.         | Res. | Res.      | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | Res. | Res. | Res. | AWHTF[3:0]    |      |      |      | Res. | Res.      | Res.       | Res.         | AWLTF[3:0]    |            |        |        |       |
|        | reset value      |             |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      | 0             | 0    | 0    | 0    |      |           |            | 0            | 0             | 0          | 0      | 0      |       |
| 0x12C  | DFSDM_FLT0AWCFR  | Res.        | Res. | Res. | Res. | Res.         | Res. | Res.      | Res. | Res. | Res. | Res. | Res. | Res.          | Res. | Res. | Res. | Res. | Res. | Res. | Res. | CLRAWHTF[3:0] |      |      |      | Res. | Res.      | Res.       | Res.         | CLRAWLTF[3:0] |            |        |        |       |
|        | reset value      |             |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      | 0             | 0    | 0    | 0    |      |           |            | 0            | 0             | 0          | 0      | 0      |       |
| 0x130  | DFSDM_FLT0EXMAX  | EXMAX[23:0] |      |      |      |              |      |           |      |      |      |      |      |               |      |      |      |      |      |      |      |               |      |      | Res. | Res. | Res.      | Res.       | EXMAXCH[1:0] |               |            |        |        |       |
|        | reset value      | 1           | 0    | 0    | 0    | 0            | 0    | 0         | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0             | 0    | 0    | 0    |      |           |            |              |               |            | 0      | 0      |       |

Table 108. DFSDM register map and reset values (continued)

| Offset        | Register name     | 31           | 30  | 29     | 28   | 27  | 26        | 25  | 24       | 23  | 22  | 21  | 20     | 19         | 18    | 17    | 16        | 15  | 14   | 13          | 12        | 11  | 10  | 9            | 8   | 7   | 6         | 5   | 4   | 3      | 2         | 1            | 0            |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|---------------|-------------------|--------------|-----|--------|------|-----|-----------|-----|----------|-----|-----|-----|--------|------------|-------|-------|-----------|-----|------|-------------|-----------|-----|-----|--------------|-----|-----|-----------|-----|-----|--------|-----------|--------------|--------------|-----------|-----|--|-----|--|-----|--|-----|--|-----|--|-----|--|-----|--|-----|--|-----|-----|-----|-----|--------------|
| 0x134         | DFSDM_FLT0EXMIN   | EXMIN[23:0]  |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           |     | Res |        | Res       |              | Res          |           | Res |  | Res |  | Res |  | Res |  | Res |  | Res |  | Res |  | Res |  | Res |     | Res |     | EXMINCH[1:0] |
|               | reset value       | 0            | 1   | 1      | 1    | 1   | 1         | 1   | 1        | 1   | 1   | 1   | 1      | 1          | 1     | 1     | 1         | 1   | 1    | 1           | 1         | 1   | 1   | 1            | 1   | 1   |           |     |     |        |           |              | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x138         | DFSDM_FLT0CNVTIMR | CNVCNT[27:0] |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           |     |     |        |           |              |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     | Res | Res | Res |              |
|               | reset value       | 0            | 0   | 0      | 0    | 0   | 0         | 0   | 0        | 0   | 0   | 0   | 0      | 0          | 0     | 0     | 0         | 0   | 0    | 0           | 0         | 0   | 0   | 0            | 0   | 0   | 0         | 0   | 0   | 0      |           |              |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x13C - 0x17C | Reserved          | Res          | Res | Res    | Res  | Res | Res       | Res | Res      | Res | Res | Res | Res    | Res        | Res   | Res   | Res       | Res | Res  | Res         | Res       | Res | Res | Res          | Res | Res | Res       | Res | Res | Res    | Res       | Res          | Res          | Res       |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x180         | DFSDM_FLT1CR1     | Res          | Res | AWFSEL | FAST | Res | Res       | Res | ROH[1:0] |     | Res | Res | RDMAEN | Res        | RSYNC | RCONT | RSW START | Res | Res  | JEXTEN[1:0] |           | Res | Res | JEXTSEL[2:0] |     | Res | Res       | Res | Res | JDMAEN | JSCAN     | JSYNC        | Res          | JSW START |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       |              |     | 0      | 0    |     |           |     | 0        | 0   |     |     | 0      |            | 0     | 0     | 0         |     |      | 0           | 0         |     |     | 0            | 0   | 0   |           |     | 0   | 0      | 0         | 0            | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x184         | DFSDM_FLT1CR2     | Res          | Res | Res    | Res  | Res | Res       | Res | Res      | Res | Res | Res | Res    | AWDCH[3:0] |       |       | Res       | Res | Res  | Res         | EXCH[3:0] |     |     | Res          | Res | Res | Res       | Res | Res | Res    | AWDIE     | ROVRIE       | JOVRIE       |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       |              |     |        |      |     |           |     |          |     |     |     |        |            | 0     | 0     | 0         | 0   |      |             |           |     | 0   | 0            | 0   | 0   |           |     |     | 0      | 0         | 0            | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x188         | DFSDM_FLT1ISR     | Res          | Res | Res    | Res  | Res | Res       | Res | Res      | Res | Res | Res | Res    | Res        | Res   | Res   | Res       | Res | RCIP | JCIP        | Res       | Res | Res | Res          | Res | Res | Res       | Res | Res | AWDF   | ROVRF     | JOVRF        | REOCF        |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       |              |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     | 0    | 0           |           |     |     |              |     |     |           |     | 0   | 0      | 0         | 0            | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x18C         | DFSDM_FLT1ICR     | Res          | Res | Res    | Res  | Res | Res       | Res | Res      | Res | Res | Res | Res    | Res        | Res   | Res   | Res       | Res | Res  | Res         | Res       | Res | Res | Res          | Res | Res | Res       | Res | Res | Res    | CLR ROVRF | CLR JOVRF    | Res          |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       |              |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           |     |     | 0      | 0         |              |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x190         | DFSDM_FLT1JCHGR   | Res          | Res | Res    | Res  | Res | Res       | Res | Res      | Res | Res | Res | Res    | Res        | Res   | Res   | Res       | Res | Res  | Res         | Res       | Res | Res | Res          | Res | Res | Res       | Res | Res | Res    | JCHG[3:0] |              |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       |              |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           |     |     | 0      | 0         | 0            | 1            |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x194         | DFSDM_FLT1FCR     | FORD[2:0]    |     | Res    | Res  | Res | FOSR[9:0] |     |          |     |     |     |        |            |       | Res   | Res       | Res | Res  | Res         | Res       | Res | Res | Res          | Res | Res | IOSR[7:0] |     |     |        |           |              |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       | 0            | 0   | 0      |      |     |           | 0   | 0        | 0   | 0   | 0   | 0      | 0          | 0     | 0     | 0         |     |      |             |           |     |     |              |     | 0   | 0         | 0   | 0   | 0      | 0         | 0            | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x198         | DFSDM_FLT1JDATAR  | JDATA[23:0]  |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           | Res | Res | Res    | Res       | JDATACH[1:0] |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       | 0            | 0   | 0      | 0    | 0   | 0         | 0   | 0        | 0   | 0   | 0   | 0      | 0          | 0     | 0     | 0         | 0   | 0    | 0           | 0         | 0   | 0   | 0            | 0   | 0   |           |     |     |        |           | 0            | 0            |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x19C         | DFSDM_FLT1RDATAR  | RDATA[23:0]  |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           | Res | Res | Res    | RPEND     | Res          | RDATACH[1:0] |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       | 0            | 0   | 0      | 0    | 0   | 0         | 0   | 0        | 0   | 0   | 0   | 0      | 0          | 0     | 0     | 0         | 0   | 0    | 0           | 0         | 0   | 0   | 0            | 0   | 0   | 0         | 0   | 0   |        |           |              | 0            | 0         |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
| 0x1A0         | DFSDM_FLT1AWHTR   | AWHT[23:0]   |     |        |      |     |           |     |          |     |     |     |        |            |       |       |           |     |      |             |           |     |     |              |     |     |           | Res | Res | Res    | Res       | BKAWH[3:0]   |              |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |
|               | reset value       | 0            | 0   | 0      | 0    | 0   | 0         | 0   | 0        | 0   | 0   | 0   | 0      | 0          | 0     | 0     | 0         | 0   | 0    | 0           | 0         | 0   | 0   | 0            | 0   | 0   |           |     |     | 0      | 0         | 0            | 0            |           |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |  |     |     |     |     |              |

Table 108. DFSDM register map and reset values (continued)

| Offset      | Register name     | 31           | 30  | 29  | 28  | 27  | 26  | 25  | 24  | 23  | 22  | 21  | 20  | 19  | 18  | 17  | 16  | 15  | 14  | 13  | 12  | 11            | 10  | 9   | 8   | 7   | 6   | 5   | 4   | 3   | 2            | 1             | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-------------|-------------------|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|---------------|-----|-----|-----|-----|-----|-----|-----|-----|--------------|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 0x1A4       | DFSDM_FLT1AWLTR   | AWLT[23:0]   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               |     |     |     |     | Res | Res | Res | Res | BKAWL[3:0]   |               |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|             | reset value       | 0            | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0             | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0            | 0             |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1A8       | DFSDM_FLT1AWSR    | Res          | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | AWHTF[3:0]    |     |     |     | Res | Res | Res | Res | Res | Res          | AWLTF[3:0]    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|             | reset value       |              |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               | 0   | 0   | 0   | 0   |     |     |     |     | 0            | 0             | 0   | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1AC       | DFSDM_FLT1AWCFR   | Res          | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | CLRAWHTF[3:0] |     |     |     | Res | Res | Res | Res | Res | Res          | CLRAWLTF[3:0] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|             | reset value       |              |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               | 0   | 0   | 0   | 0   |     |     |     |     | 0            | 0             | 0   | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1B0       | DFSDM_FLT1EXMAX   | EXMAX[23:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               |     |     |     |     | Res | Res | Res | Res | EXMAXCH[1:0] |               |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|             | reset value       | 1            | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0             | 0   | 0   | 0   | 0   |     |     |     |     |              |               | 0   | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1B4       | DFSDM_FLT1EXMIN   | EXMIN[23:0]  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               |     |     |     |     | Res | Res | Res | Res | EXMINCH[1:0] |               |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|             | reset value       | 0            | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1             | 1   | 1   | 1   | 1   |     |     |     |     |              |               | 0   | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1B8       | DFSDM_FLT1CNVTIMR | CNVCNT[27:0] |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |               |     |     |     |     |     |     |     |     | Res          | Res           | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res |
|             | reset value       | 0            | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0             | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0            | 0             | 0   | 0   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| 0x1BC-0x2FC | Reserved          | Res          | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res | Res           | Res | Res | Res | Res | Res | Res | Res | Res | Res          | Res           | Res | Res |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

Refer to [Section 2.2 on page 70](#) for the register boundary addresses.